

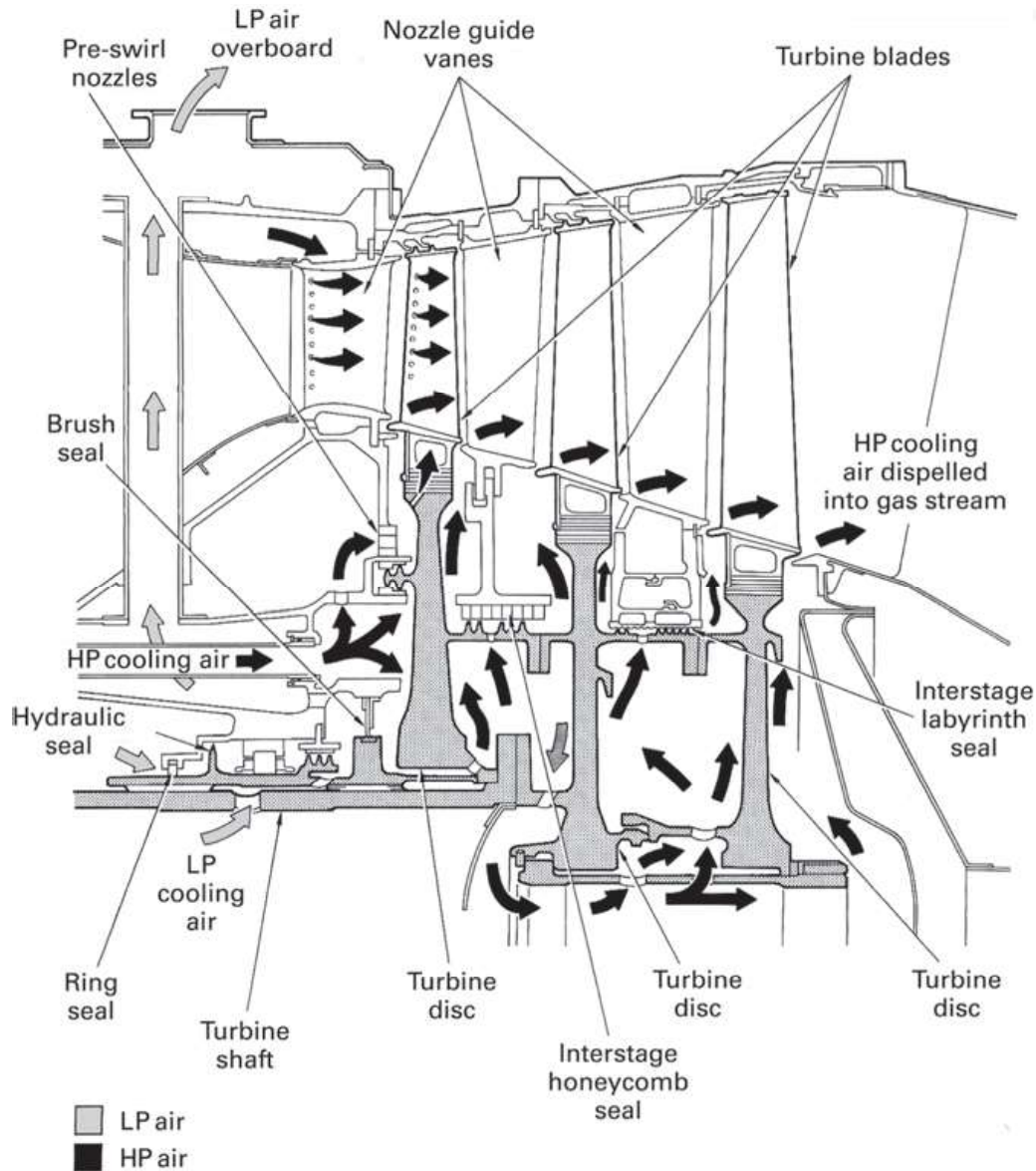


FIG. 2.19 Cooling air distribution [courtesy Rolls-Royce plc]

2.3 Design point performance calculations

Before assessing the effect of component losses on the general performance curves for the various cycles considered in section 2.1, it is necessary to outline the method of calculating the performance in any particular case for specified values of the design parameters. These parameters will include the compressor pressure ratio, turbine inlet temperature (TIT), component efficiencies and pressure losses.

Several examples will be used to show how these effects are incorporated in realistic cycle calculations. For the first example we consider a single-shaft unit with a heat-exchanger, using modest values of pressure ratio and turbine inlet temperature, and typical isentropic efficiencies. A second example shows how to calculate the performance of a simple cycle of high pressure ratio and turbine temperature, which incorporates a free power turbine. Finally, we will evaluate the performance of an advanced cycle using reheat, suitable for use in both simple and combined cycle applications; this example will illustrate the use of polytropic efficiencies.

EXAMPLE 2.1

Determine the specific work output, specific fuel consumption and cycle efficiency for a heat-exchange cycle, Fig. 2.20, having the following specification:

Compressor pressure ratio	4.0
Turbine inlet temperature	1100 K
Isentropic efficiency of compressor, η_c	0.85
Isentropic efficiency of turbine, η_t	0.87
Mechanical transmission efficiency, η_m	0.99
Combustion efficiency, η_b	0.98
Heat-exchanger effectiveness	0.80
Pressure losses:	
Combustion chamber, Δp_b	2% comp. deliv. press.
Heat-exchanger air side, Δp_{ha}	3% comp. deliv. press.
Heat-exchanger gas side, Δp_{hg}	0.04 bar
Ambient conditions, p_a, T_a	1 bar, 288 K

Since $T_{01} = T_a$ and $p_{01} = p_a$, and $\gamma = 1.4$, the temperature equivalent of the compressor work from equation (2.11) is

$$\begin{aligned}
 T_{02} - T_a &= \frac{T_a}{\eta_c} \left[\left(\frac{p_{02}}{p_a} \right)^{(\gamma-1)/\gamma} - 1 \right] \\
 &= \frac{288}{0.85} [4^{1/3.5} - 1] = 164.7 \text{ K}
 \end{aligned}$$

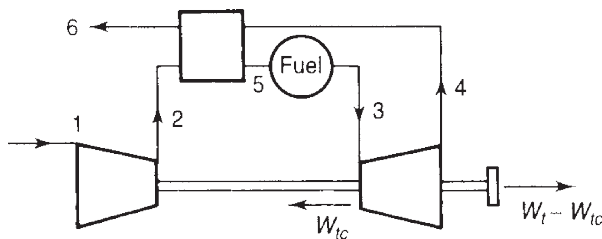


FIG. 2.20 Heat-exchange cycle

Turbine work required to drive compressor per unit mass flow is

$$W_{tc} = \frac{c_{pa}(T_{02} - T_a)}{\eta_m} = \frac{1.005 \times 164.7}{0.99} = 167.2 \text{ kJ/kg}$$

$$p_{03} = p_{02} \left(1 - \frac{\Delta p_b}{p_{02}} - \frac{\Delta p_{ha}}{p_{02}} \right) = 4.0(1 - 0.02 - 0.03) = 3.8 \text{ bar}$$

$$p_{04} = p_a + \Delta p_{hg} = 1.04 \text{ bar, and hence } p_{03}/p_{04} = 3.654$$

Since $\gamma = 1.333$ for the expanding gases, the temperature equivalent of the total turbine work from equation (2.13) is

$$\begin{aligned} T_{03} - T_{04} &= \eta_t T_{03} \left[1 - \left(\frac{1}{p_{03}/p_{04}} \right)^{(\gamma-1)/\gamma} \right] \\ &= 0.87 \times 1100 \left[1 - \left(\frac{1}{3.654} \right)^{1/4} \right] = 264.8 \text{ K} \end{aligned}$$

Total turbine work per unit mass flow is

$$W_t = c_{pg}(T_{03} - T_{04}) = 1.148 \times 264.8 = 304.0 \text{ kJ/kg}$$

Remembering that the mass flow is to be assumed the same throughout the unit, the specific work output is simply

$$W_t - W_{tc} = 304 - 167.2 = 136.8 \text{ kJ/kg (or kW s/kg)}^\dagger$$

(It follows that for a 1000 kW plant an air mass flow of 7.3 kg/s would be required.) To find the fuel/air ratio we must first calculate the combustion temperature rise ($T_{03} - T_{05}$).

From equation (2.22),

$$\text{heat-exchanger effectiveness} = 0.80 = \frac{T_{05} - T_{02}}{T_{04} - T_{02}}$$

$$T_{02} = 164.7 + 288 = 452.7 \text{ K, and } T_{04} = 1100 - 264.8 = 835.2 \text{ K}$$

Hence

$$T_{05} = 0.80 \times 382.5 + 452.7 = 758.7 \text{ K}$$

From Fig. 2.17, for a combustion chamber inlet air temperature of 759 K and a combustion temperature rise of $(1100 - 759) = 341 \text{ K}$, the theoretical fuel/air ratio required is 0.0094 and thus

$$f = \frac{\text{theoretical } f}{\eta_b} = \frac{0.0094}{0.98} = 0.0096$$

[†] Note: 1 kW s/kg $\equiv$ 0.6083 hp s/lb.

The specific fuel consumption is therefore

$$SFC = \frac{f}{W_t - W_{tc}} = \frac{3600 \times 0.0096}{136.8} = 0.253 \text{ kg/kW h}^\dagger$$

Finally, the cycle efficiency is

$$\eta = \frac{3600}{SFC \times Q_{\text{net},p}} = \frac{3600}{0.253 \times 43\,100} = 0.331$$

A modern engine would operate at a significantly higher turbine inlet temperature, and this in turn would require the use of a higher pressure ratio. Repeating the above calculation with a pressure ratio of 10.0 and a temperature of 1450 K results in an increase of specific output by 115 per cent and the cycle efficiency is raised to 39.1 per cent; this efficiency is close to that of the Solar Mercury, which operates at similar cycle conditions.

EXAMPLE 2.2

Determine the specific work output, specific fuel consumption and cycle efficiency for a simple cycle gas turbine with a free power turbine (Fig. 2.21) given the following specification:

Compressor pressure ratio	12.0
Turbine inlet temperature	1350 K
Isentropic efficiency of compressor, η_c	0.86
Isentropic efficiency of each turbine, η_t	0.89
Mechanical efficiency of each shaft, η_m	0.99
Combustion efficiency	0.99
Combustion chamber pressure loss	6% comp. deliv. press.
Exhaust pressure loss	0.03 bar
Ambient conditions, p_a, T_a	1 bar, 288 K

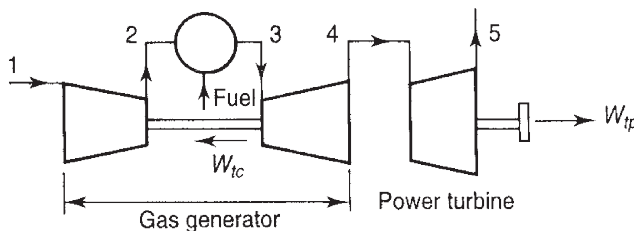


FIG. 2.21 Free turbine unit

[†] Note: 1 kg/kW h $\equiv$ 1.644 lb/hp h.

Proceeding as in the previous example,

$$T_{02} - T_{01} = \frac{288}{0.86} [12^{1/3.5} - 1] = 346.3 \text{ K}$$

$$W_{tc} = \frac{1.005 \times 346.3}{0.99} = 351.5 \text{ kJ/kg}$$

$$p_{02} = 12.0(1 - 0.06) = 11.28 \text{ bar}$$

The intermediate pressure between the two turbines, p_{04} , is unknown but can be determined from the fact that the compressor turbine produces just sufficient work to drive the compressor. The temperature equivalent of the compressor turbine work is therefore

$$T_{03} - T_{04} = \frac{W_{tc}}{c_{pg}} = \frac{351.5}{1.148} = 306.2 \text{ K}$$

The corresponding pressure ratio can be found using equation (2.12),

$$T_{03} - T_{04} = \eta_t T_{03} \left[1 - \left(\frac{1}{p_{03}/p_{04}} \right)^{\gamma-1/\gamma} \right]$$

$$306.2 = 0.89 \times 1350 \left[1 - \left(\frac{1}{p_{03}/p_{04}} \right)^{0.25} \right]$$

$$\frac{p_{03}}{p_{04}} = 3.243$$

$$T_{04} = 1350 - 306.2 = 1043.8 \text{ K}$$

The pressure at entry to the power turbine, p_{04} , is then found to be $11.28/3.243 = 3.478$ bar and the power turbine pressure ratio is $3.478/(1 + 0.03) = 3.377$.

The temperature drop in the power turbine can now be obtained from equation (2.12),

$$T_{04} - T_{05} = 0.89 \times 1043.8 \left[1 - \left(\frac{1}{3.377} \right)^{0.25} \right] = 243.7 \text{ K}$$

and the specific work output, i.e. power turbine work per unit air mass flow, is

$$W_{tp} = C_{pg}(T_{04} - T_{05})\eta_m$$

$$W_{tp} = 1.148(243.7)0.99 = 277.0 \text{ kJ/kg (or kW/kg)}$$

The compressor delivery temperature is $288 + 346.3 = 634.3$ K and the combustion temperature rise is $1350 - 634.3 = 715.7$ K; from Fig. 2.17, the theoretical

fuel/air ratio required is 0.0202 giving an actual fuel/air ratio of $0.0202/0.99 = 0.0204$.

The *SFC* and cycle efficiency, η , are then given by

$$SFC = \frac{f}{W_{tp}} = \frac{3600 \times 0.0204}{277.9} = 0.265 \text{ kg/kW h}$$

$$\eta = \frac{3600}{0.265 \times 43\,100} = 0.315$$

It should be noted that the cycle calculations have been carried out as above to determine the *overall* performance. It is important to realize, however, that they also provide information that is needed by other groups such as the aerodynamic and control design groups, as discussed in section 1.9. The temperature at entry to the power turbine, T_{04} , for example, may be required as a control parameter to prevent operation above the metallurgical limiting temperature of the compressor turbine. The exhaust gas temperature (*EGT*), T_{05} , would be important if the gas turbine were to be considered for combined cycle or cogeneration plant, which were first mentioned in Chapter 1, and will also be discussed in section 2.5. For the cycle of Example 2.2, $T_{05} = 1043.8 - 243.7 = 800.1 \text{ K}$ or 527°C , which is suitable for use with a waste heat boiler. When thinking of combined cycle plant, a higher *TIT* might be desirable because there would be a consequential increase in *EGT*, permitting the use of a higher steam temperature and a more efficient steam cycle. If the cycle pressure ratio were increased to increase the efficiency of the gas cycle, however, the *EGT* would be decreased resulting in a lower steam cycle efficiency. The next example will illustrate how the need to make a gas turbine suitable for more than one application can affect the choice of cycle parameters.

EXAMPLE 2.3

Consider the design of a high pressure ratio, single-shaft cycle with reheat at some point in the expansion when used *either* as a separate unit, *or* as part of a combined cycle. The power required is 240 MW at 288 K and 1.01 bar.

Compressor pressure ratio	30
Polytropic efficiency (compressor and turbines)	0.89
Turbine inlet temperature (both turbines)	1525 K
$\Delta p/p_{02}$ (1st combustor)	0.02
$\Delta p/p_{04}$ (2nd combustor)	0.04
Exhaust pressure	1.02 bar

The plant is shown in Fig. 2.22. A heat-exchanger is not used because it would result in an exhaust temperature that would be too low for use with a high-efficiency steam cycle.

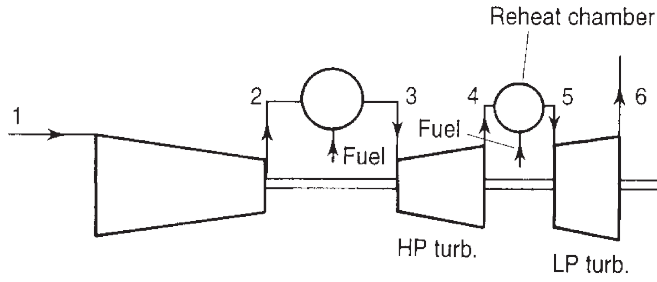


FIG. 2.22 Reheat cycle

To simplify the calculation it will be assumed that the mass flow is constant throughout, ignoring the effect of the substantial cooling bleeds that would be required with the high turbine inlet temperatures specified. The reheat pressure is not specified, but as a starting point it is reasonable to use the value giving the same pressure ratio in each turbine. (As shown in section 2.1, this division of the expansion leads to equal work in each turbine and a maximum net work output for the *ideal* reheat cycle.)

It is convenient to start by evaluating $(n - 1)/n$ for the polytropic compression and expansion. Referring to equations (2.19) and (2.20), we have

$$\text{for compression: } \frac{n - 1}{n} = \frac{1}{\eta_{\infty c}} \left(\frac{\gamma - 1}{\gamma} \right) = \frac{1}{0.89} \left(\frac{0.4}{1.4} \right) = 0.3210$$

$$\text{for expansion: } \frac{n - 1}{n} = \eta_{\infty t} \left(\frac{\gamma - 1}{\gamma} \right) = 0.89 \left(\frac{0.333}{1.333} \right) = 0.2223$$

Making the usual assumption that $p_{01} = p_a$ and $T_{01} = T_a$, we have

$$\frac{T_{02}}{T_{01}} = (30)^{0.3210} \quad T_{02} = 858.1 \text{ K}$$

$$T_{02} - T_{01} = 570.1 \text{ K}$$

$$p_{02} = 30 \times 1.01 = 30.3 \text{ bar}$$

$$p_{03} = 30.3(1.00 - 0.02) = 29.69 \text{ bar}$$

$$p_{06} = 1.02 \text{ bar, so } \frac{p_{03}}{p_{06}} = 29.11$$

Theoretically, the optimum pressure ratio for each turbine would be $\sqrt[4]{(29.11)} = 5.395$. The 4 per cent pressure loss in the reheat combustor has to be considered, so a value of 5.3 for p_{03}/p_{04} could be assumed. Then

$$\frac{T_{03}}{T_{04}} = (5.3)^{0.2223} \quad T_{04} = 1052.6 \text{ K}$$

$$p_{04} = 29.69/5.3 = 5.602 \text{ bar}$$

$$p_{05} = 5.602(1.00 - 0.04) = 5.378 \text{ bar}$$

$$p_{05}/p_{06} = 5.378/1.02 = 5.272$$

$$\frac{T_{05}}{T_{06}} = (5.272)^{0.2223} \quad T_{06} = 1053.8 \text{ K}$$

Assuming unit flow of 1.0 kg/s, and a mechanical efficiency of 0.99,

$$\begin{aligned} \text{Turbine output, } W_t &= 1.0 \times 1.148[(1525 - 1052.6) \\ &\quad + (1525 - 1053.8)] \times 0.99 \\ &= 1072.3 \text{ kJ/kg} \end{aligned}$$

$$\begin{aligned} \text{Compressor input, } W_c &= 1.0 \times 1.005 \times 570.1 \\ &= 573.0 \text{ kJ/kg} \end{aligned}$$

$$\text{Net work output, } W_N = 1072.3 - 573.0 = 499.3 \text{ kJ/kg}$$

Flow required for 240 MW is given by

$$m = \frac{240\,000}{499.3} = 480.6 \text{ kg/s}$$

The combustion temperature rise in the first combustor is $(1525 - 858) = 667 \text{ K}$, and with an inlet temperature of 858 K the fuel/air ratio is found from Fig. 2.17 to be 0.0197. For the second combustor the temperature rise is $(1525 - 1052.6) = 472.4 \text{ K}$ giving a fuel/air ratio of 0.0142. The total fuel/air ratio is then

$$f = \frac{0.0197 + 0.0142}{0.99} = 0.0342$$

and the thermal efficiency is given by

$$\eta = \frac{499.3}{0.0342 \times 43\,100} = 33.9 \text{ per cent}$$

This is a reasonable efficiency for simple cycle operation, and the specific output is excellent. Examination of the temperature at turbine exit, however, shows that $T_{06} = 1053.8 \text{ K}$ or 780.8°C . This temperature is too high for efficient use in a combined cycle plant. A reheat steam cycle using conventional steam temperatures of about $550\text{--}575^\circ\text{C}$ would require a turbine exit temperature of about 600°C .

The turbine exit temperature could be reduced by increasing the reheat pressure, and if the calculations are repeated for a range of reheat pressure the results shown in Fig. 2.23 are obtained. It can be seen that a reheat pressure of 13 bar gives an *EGT* of 605°C ; the specific output is about 10 per cent lower than the optimum value, but the thermal efficiency is substantially improved to 37.7 per cent. Further increases in reheat pressure would give slightly higher efficiencies, but the *EGT* would be reduced below 600°C resulting in a less efficient steam cycle. With a reheat pressure of 13 bar, the first turbine has a pressure ratio of 2.284 while the second has a pressure ratio of 12.23, differing markedly from the equal pressure

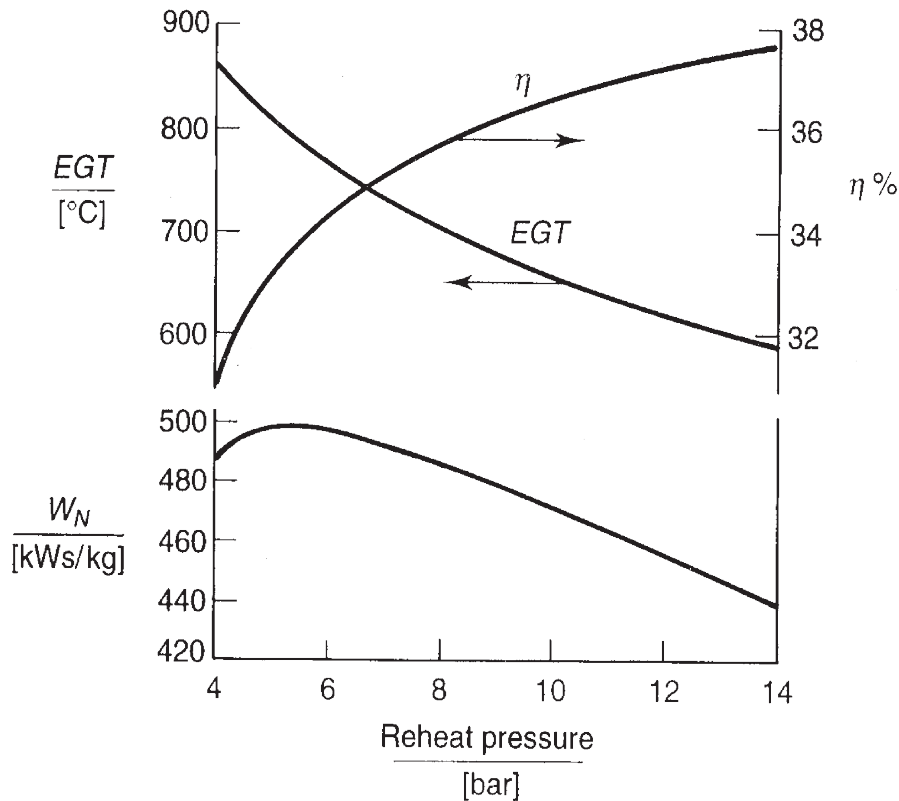


FIG. 2.23 Effect of varying reheat pressure

ratios we assumed at the outset. This example illustrates some of the problems that arise when a gas turbine must be designed for more than one application.

Figure 2.24 compares the cross-section of the Alstom GT26 reheat (sequential combustion) gas turbine with the conventional ABB GT13E2 gas turbine. It can be seen that the use of reheat provided an increase from 165 to 241 MW for the same footprint, primarily due to the increase in specific power. By 2015, the ratings had increased to 203 and 345 MW respectively as a result of continuous development. The increase in power density proved valuable in re-powering applications where the existing site was cramped for space. Alstom was taken over by GE in 2015 and the GT 13E2 is still offered by GE but the GT26 was transferred to Ansaldo Energia.

It can be seen that the GT13E2, on the lower half of the centre line, has a five-stage turbine. The GT26 also has five stages of turbine, with a single-stage HP turbine located between the two combustors and a four-stage LP turbine, appropriate for the reheat pressure required for an *EGT* of 600 C.

The table in Chapter 1 showing the long-term development of the W501G shows that it always has used a four-stage turbine, and this can be seen in the picture of the latest 501G (or Siemens SGT6-5000F). For many years GE used three-stage turbines, with higher aerodynamic loading but lower manufacturing cost; at current higher values of pressure ratio, GE have now moved to a four-stage turbine.

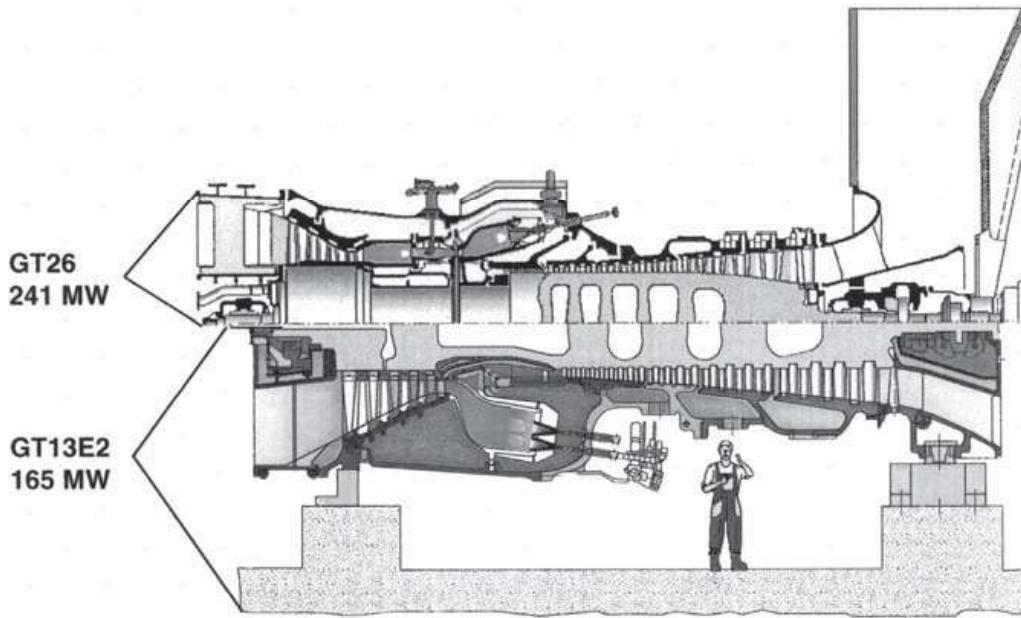


FIG. 2.24 Reheat and conventional gas turbines Source [courtesy Alstom Power Ltd]

2.4 Comparative performance of practical cycles

The large number of variables involved make it impractical to derive algebraic expressions for the specific output and efficiency of real cycles. The methods derived in the previous section can readily be programmed to evaluate performance over a range of cycle conditions.

Some performance curves will now be presented to show the main differences between practical and ideal cycles, and the relative importance of some of the parameters. The curves are definitely *not* a comprehensive set from which designers can make a choice of cycle for a particular application. To emphasize that too much importance should not be attached to the values of specific output and efficiency, the full specification of the parameters has not been given: it is sufficient to note that those parameters which are not specified on the curves are kept constant. All the curves use compressor pressure ratio r_c as abscissa, the turbine pressure ratio being less than r_c by virtue of the pressure losses. The cycle efficiency has been evaluated to facilitate comparison with the ideal curves of section 2.1. In practice it is usual to quote *SFC* rather than efficiency, not only because its definition is unambiguous, but also because it provides both a direct indication of fuel consumption and a measure of cycle efficiency to which it is inversely proportional.

Simple gas turbine cycle

When component losses are taken into account the efficiency of the simple cycle becomes dependent on both the pressure ratio and the maximum cycle temperature T_{03} , as shown in Fig. 2.25. For each temperature the efficiency has a maximum value

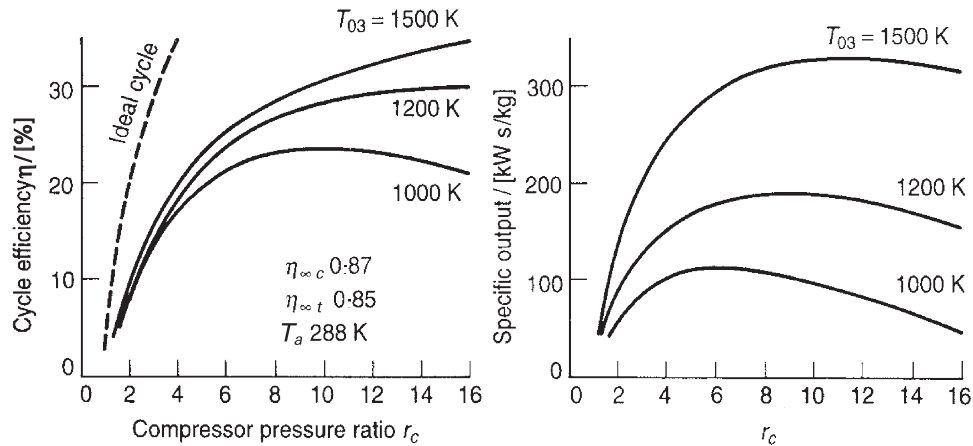


FIG. 2.25 Cycle efficiency and specific output of simple gas turbine

at a particular pressure ratio, which increases with maximum cycle temperature. The fall in efficiency at higher pressure ratios is due to the fact that the reduction in fuel supply to give the required turbine inlet temperature, resulting from the higher compressor delivery temperature, is outweighed by the increased work necessary to drive the compressor. The optimum pressure ratio for maximum efficiency differs from that for maximum specific output, but it should be noted that the curves are fairly flat near the maxima. Normally the pressure ratio would be selected on the basis of efficiency, and the chosen pressure ratio may be slightly lower than the optimum value. The choice of pressure ratio would be affected by issues such as avoiding the use of excessively small blades at the HP end of the compressor, the number of compressor and turbine stages required, and whirling speed and bearing problems associated with the length of the compressor-turbine combination. The choice of pressure ratio and turbine inlet temperature would also vary with the power output; low-power engines would require lower pressure ratio because of blade size limitations and also lower turbine temperatures because of the difficulty of cooling very small blades.

The advantage of using as high a value of T_{03} as possible, and the need to use a higher pressure ratio to take advantage of the higher permissible temperature, is evident from Fig. 2.25. It can be seen that the effect of increasing turbine inlet temperature is much greater for the specific output than for the efficiency. The efficiency increases with T_{03} because the component losses become relatively less important as the ratio of positive turbine work to negative compressor work increases, although the gain in efficiency would be reduced if T_{03} were increased to a level where blade cooling introduced additional losses. The marked gain in specific power with increase of maximum cycle temperature results in a significant reduction in plant size for a given power.

Heat-exchange (or regenerative) cycle

As far as the specific work output is concerned, the addition of a heat-exchanger merely causes a slight reduction due to the additional pressure losses; the curves retain the same form as in Fig. 2.25. The efficiency curves are very different, as

shown in Fig. 2.26. Heat-exchange increases the efficiency substantially and markedly reduces the optimum pressure ratio for maximum efficiency. Figure 2.26 shows that the effect of T_{03} on efficiency is much greater than for the simple cycle. This is an important feature of the heat-exchange cycle, because maximum cycle temperatures have increased at a rate of about 10 K per year and there is every hope that this will continue. It should be noted, however, that increasing the cycle temperature also increases the temperature at turbine exit, i.e. at entry to the heat-exchanger. With stainless steel the maximum heat-exchanger temperature is limited to about 900 K, and with more expensive chromium-nickel alloys this may be raised to about 1100 K. Figure 2.26 shows that the optimum pressure ratio for maximum efficiency increases as T_{03} is increased. Our study of the ideal heat-exchange cycle suggested that no such increase in pressure ratio was required; this was the conclusion under the heading 'Heat-exchanger cycle' in section 2.1 which it was stated would have to be modified. Nevertheless, it remains true that no very high pressure ratio is used with heat-exchangers, which are subjected to high internal pressures and also severe thermal stresses during start-up and shut down.

The dashed curves have been added to Fig. 2.26 to show the effect of heat-exchanger effectiveness. An increase in effectiveness has an appreciable effect on cycle efficiency, and it also reduces the value of the optimum pressure ratio; once again, the curves are fairly flat around the optimum. A modern heat-exchange cycle may operate at a pressure ratio of 9–10 and a turbine inlet temperature of around 1500 K.

A more useful method of presenting performance characteristics is to plot the variation of specific fuel consumption and specific output on a single figure for a range of values of pressure ratio and turbine inlet temperature. Examples for realistic values of polytropic efficiency, pressure losses and heat-exchanger effectiveness are shown in Figs. 2.27(a) and 2.27(b) for simple and heat-exchange cycles respectively. The marked effect of increasing T_{03} on specific output in both cases

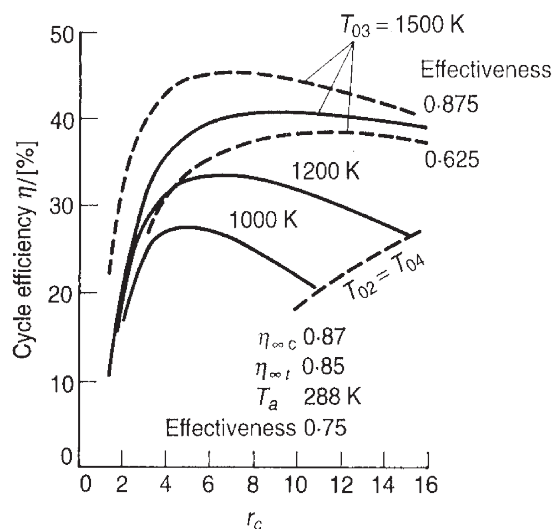


FIG. 2.26 Heat-exchange cycle

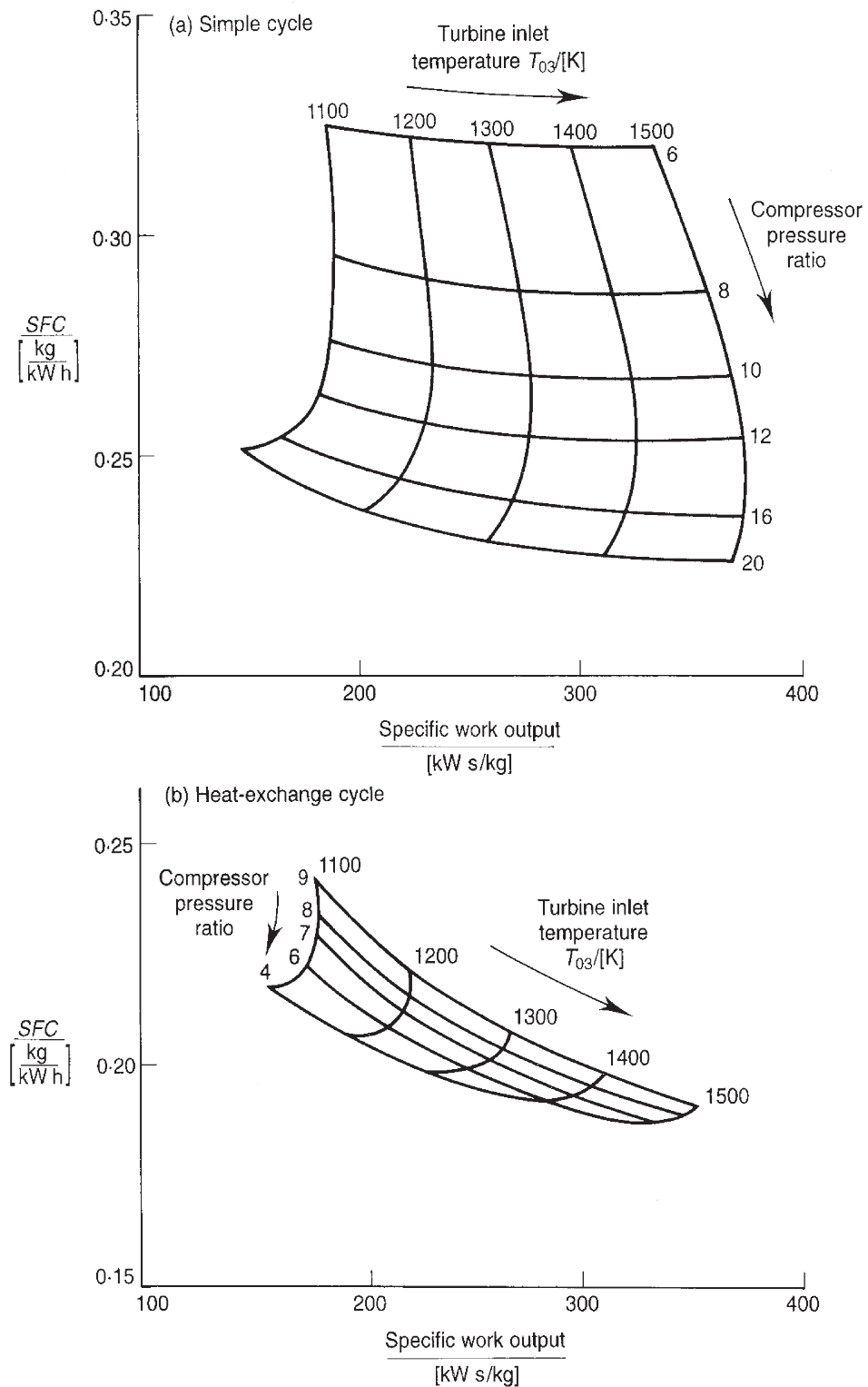


FIG. 2.27 Cycle performance curves

is clearly evident. Such plots also contrast clearly the small effect of T_{03} on SFC for the simple cycle with the much greater effect when a heat-exchanger is used.

Heat-exchange (regenerative) cycle with reheat or intercooling

Complex cycles including regeneration, reheat and intercooling were investigated in great detail in the early days of gas turbine development, when pressure ratios and turbine inlet temperatures were very low by current standards. These cycles were not further considered for decades, but both reheat and intercooling re-emerged at the end of the 20th century. Figure 2.28 shows that the combination of heat-exchange and reheat improves the specific output considerably without loss of efficiency (cf. Figs. 2.28 and 2.25/2.26). The curves of Fig. 2.28 are based on the assumption that the gas is reheated to the maximum cycle temperature at the point in the expansion giving equal pressure ratios for the two turbines. The gain in efficiency due to reheat obtained with the ideal cycle is not realized in practice, partly because of the additional pressure loss in the reheat chamber, but primarily because the heat-exchanger effectiveness is well short of unity, and the additional energy in the exhaust is not fully recovered.

Reheat was reintroduced to the market by ABB in the late 1990s; this cycle did not use heat-exchange, the idea being to use the high exhaust temperature in an HRSG to raise steam at advanced cycle conditions. ABB referred to the reheat process as *sequential combustion*, and the use of reheat allowed the use of a high-pressure ratio cycle which still maintained a high EGT ; this made the cycle competitive both for simple cycle and combined cycle applications. At this time it would be instructive to review Example 2.3, which shows how the reheat pressure selected was based on the required EGT (about 600°C) rather than maximizing the specific work output.

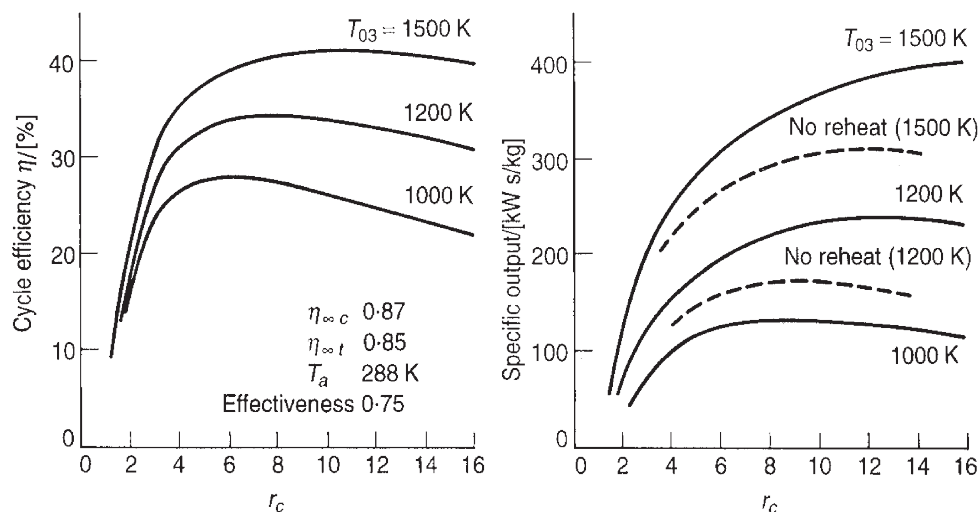


FIG. 2.28 Cycle with heat-exchange and reheat

When intercooling is incorporated in a practical cycle, even allowing for the additional pressure losses, there is both a marked increase in specific output and a worthwhile improvement in efficiency. Intercoolers tend to be bulky, and if they require cooling water the self-contained nature of the gas turbine is lost; in arid regions or areas where extreme cold is experienced for long periods, causing freezing of the water supply, intercooling is clearly not viable. The intercooled regenerative cycle (ICR) is attractive for naval applications, where cooling water (i.e. the ocean) is readily available. The main advantage of the ICR cycle, however, is its greatly improved part-load performance, which will be further discussed in Chapter 10.

GE has recently brought to the market the LMS100 engine, a 100 MW unit with a thermal efficiency in excess of 40 per cent. This uses intercooling without heat exchange, and is aimed particularly at peaking applications where running hours are relatively short but the ability to come on load rapidly is very important. Most conventional peaking units are simple cycle configurations, with lower efficiencies. The LMS100 can be supplied with either water-cooled or air-cooled intercooling; the air-cooled versions would require substantial auxiliary power to drive the fans, but would permit the use of the engine in areas where water was not available.

We may conclude this section by emphasizing that in practice most gas turbines utilize a high-pressure ratio simple cycle. In the 1950s and 1960s a number of heat-exchange cycles were built, but the only widespread use was in the Honeywell AGT 1500 in the M1A1 tank for the US Army. It can now be seen that modern engines such as the Solar Mercury, Alstom GT26, Rolls-Royce WR21 (ICR) and GE LMS100 are bringing some of these old ideas back. Studies suggest that efficiencies of over 60 per cent may be achieved by combining intercooling, heat-exchange and reheat [Ref. (5)].

2.5 Combined cycles and cogeneration schemes

In the gas turbine, practically all the energy not converted to shaft power is available in the exhaust gases for other uses. The only limitation is that the final exhaust temperature, i.e. the stack temperature, should not be reduced below the dew point to avoid corrosion problems arising from sulphur in the fuel. The exhaust heat may be used in a variety of ways. If it is wholly used to produce steam in a waste heat boiler or HRSG for a steam turbine, with the object of augmenting the shaft power produced, the system is referred to as a *combined gas/steam cycle power plant* or simply a *combined cycle plant* (Fig. 1.3). Alternatively, the exhaust heat may be used to produce hot water or steam for district or factory heating, hot gas or steam for some chemical process, hot gas for distillation plant, or steam for operating an absorption refrigerator in water chilling or air-conditioning plant. The shaft power will normally be used to produce electricity. In such circumstances the system is referred to as a *cogeneration* or CHP plant (Fig. 1.18). We shall consider briefly the main characteristics of these two types of system in turn.

Combined cycle plant

The optimization of binary cycles is too complex to be discussed in detail in this book, but the more important decisions that have to be taken when designing combined cycle plant can be described in broad terms.

Figure 2.29(a) shows the gas and steam conditions in the boiler on a T - H diagram. The enthalpy rise between feed water inlet and steam outlet must equal the enthalpy drop of the exhaust gases in the HRSG, and the pinch point and terminal temperature differences cannot be less than about 20°C if the boiler is to be of economic size. It follows that a reduction in gas turbine exhaust temperature will lead to a reduction in the steam pressure that can be used for the steam cycle. In the combined cycle, therefore, selection of a higher compressor pressure ratio to improve the gas turbine efficiency may lead to a fall in steam cycle efficiency and a reduction in overall thermal efficiency. In practice, heavy industrial gas turbines must be acceptable in both simple and combined cycle markets and it appears that a pressure ratio of around 15, giving an EGT of between 550 and 600°C , is suitable for both. Aircraft derivative engines with design pressure ratios of 25–35 yield lower values of EGT of about 450°C which result in the use of less efficient steam cycles.

Combined cycle plant is used for large base-load generating stations and the overall thermal efficiency, while very important, is not the ultimate criterion, which is the cost of a unit of electricity sent out. This will depend on both the efficiency and the capital cost of the plant. For example, if a dual-pressure steam cycle were used, as in the plant shown in Fig. 1.3, a higher efficiency would be obtained because the average temperature at which heat is transferred to the steam is increased. Figure 2.29(b) will make this clear. The additional complication would certainly increase the cost of the boiler and steam turbine, and a detailed study would be required to see if this would lead to a reduction in the cost of

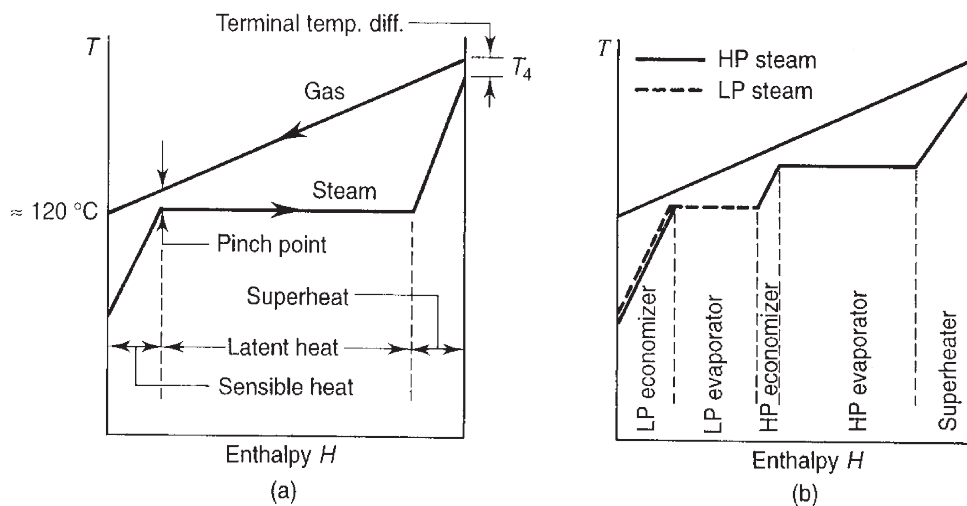


FIG. 2.29 T - H diagrams for single- and dual-pressure combined cycle plant

electricity produced. The results of such a study can be found in Ref. (6). Dual-pressure cycles have, in fact, been widely used; with modern gas turbines yielding exhaust gas temperatures close to 600°C, triple-pressure cycles with reheat have been found to be economic and are now commonly used.

Two other decisions have to be made at an early stage in the design process. There is the possibility of associating more than one gas turbine with a single HRSG and steam turbine. Such an arrangement would permit each gas turbine to be fitted with its own diverter valve and stack so that any particular gas turbine could be shut down for maintenance. A second option requiring a decision is whether or not to arrange for extra fuel to be burnt in the HRSG, referred to as *supplementary firing*. This could be used for increasing the peak load for short periods, but would increase the capital cost of the HRSG substantially. The exhaust gases from the supplementary firing do not pass through a turbine, which opens the possibility of burning heavy oil or coal in the HRSG. This is usually ruled out by the logistic difficulty and cost of supplying a large plant with more than one fuel.

We will conclude this sub-section by showing how to estimate the improvement in performance resulting from the addition of an exhaust-heated Rankine cycle to a gas cycle. Only the simple case of a single-pressure superheat cycle (Fig. 2.29(a)) will be dealt with. The heat transferred from the exhaust gas is $m_g c_{pg}(T_4 - T_{\text{stack}})$ where T_4 is the temperature at exit from the turbine. This is equal to $m_s(h - h_w)$ where m_s is the steam flow and h and h_w are the specific enthalpies at the steam turbine inlet and HRSG inlet respectively. The superheat temperature is fixed by T_4 and the terminal temperature difference, and the enthalpy will depend on the pressure. The pinch point temperature, T_p , is fixed by the pinch point temperature difference and the saturation temperature of the steam. The mass flow of steam is obtained from

$$m_s(h - h_f) = m_g c_{pg}(T_4 - T_p)$$

where h_f is the saturated liquid enthalpy. The stack temperature can then be determined from

$$m_g c_{pg}(T_p - T_{\text{stack}}) = m_s(h_f - h_w)$$

The total power from the combined cycle is $W_{gt} + W_{st}$ but the heat input (from the fuel burned in the gas turbine) is unchanged. The overall efficiency is then given by

$$\eta = \frac{W_{gt} + W_{st}}{Q}$$

It is useful to express this in terms of the gas turbine and steam turbine efficiencies to illustrate their individual effects. $W_{st} = \eta_{st} Q_{st}$, where Q_{st} is the notional heat supplied to the steam turbine:

$$Q_{st} = m_g C_{pg}(T_4 - T_{\text{stack}})$$

This, in turn, can be expressed in terms of the heat rejection from the gas turbine assuming that the turbine exhaust gas at T_4 is cooled without useful energy extraction from T_{stack} to the ambient temperature T_a , i.e. $m_g C_{pg}(T_4 - T_a)$ or $Q(1 - \eta_{gt})$. Thus

$$Q_{st} = m_g C_{pg}(T_4 - T_a) \frac{(T_4 - T_{\text{stack}})}{(T_4 - T_a)} = Q(1 - \eta_{gt}) \frac{(T_4 - T_{\text{stack}})}{(T_4 - T_a)}$$

The overall efficiency is then given by

$$\eta = \frac{W_{gt}}{Q} + \frac{W_{st}}{Q} = \frac{W_{gt}}{Q} + \frac{\eta_{st} Q_{st}}{Q}$$

or

$$\eta = \eta_{gt} + \eta_{st}(1 - \eta_{gt}) \frac{(T_4 - T_{\text{stack}})}{(T_4 - T_a)}$$

It can now be seen that the overall efficiency is influenced by both the gas turbine exhaust temperature and the stack temperature. A typical modern gas turbine used in combined cycle applications would have an exhaust gas temperature of around 600°C and a thermal efficiency of about 34 per cent. The stack temperature could be around 140°C if liquid fuel were being burned, or 120°C when the fuel was natural gas which has a very low sulphur content. A single-pressure steam cycle might give around 32 per cent thermal efficiency. Hence

$$\begin{aligned} \eta &= 0.34 + 0.32(1 - 0.34) \left[\frac{600 - 120}{600 - 15} \right] \\ &= 0.513 \end{aligned}$$

Using a more complex steam cycle with $\eta_{st} = 0.36$, the overall efficiency would be raised to 53.5 per cent, which is typical of modern combined cycles.

Cogeneration plant

In a cogeneration plant, such as that in Fig. 1.18, the steam generated in the HRSGs is used for several different purposes such as a steam turbine drive for a centrifugal water chiller, a heat source for absorption chillers, and to meet process and heating requirements. Each user, such as a hospital or paper making factory, will have different requirements for steam, and even though the same gas turbine may be suitable for a variety of cogeneration plant, the HRSGs would be tailored to the specific requirements of the user.

A major problem in the thermodynamic design of a cogeneration plant is the balancing of the electric power and steam loads. The steam requirement, for example, will depend on the heating and cooling loads which may be subject to major seasonal variations: the electrical demand may peak in either winter or summer

depending on the geographical location. In some cases, the system may be designed to meet the primary heating load and the gas turbine selected may provide significantly more power than demanded by the plant; in this case, surplus electricity would be exported to the local utility. Another possibility is for electricity generation to be the prime requirement, with steam sold as a by-product to an industrial complex for process use. The economics and feasibility of a cogeneration plant are closely related to the commercial arrangements that can be made with the local electric utility or the steam host. It would be necessary, for example, to negotiate rates at which the utility would purchase power and also to ensure that back-up power would be available during periods when the gas turbine was out of service; these could be planned outages, for maintenance, or forced outages resulting from system failure. Such considerations are crucial when the cogeneration plant forms the basis of a district heating system.

When designing a cogeneration plant care must be taken in specifying the ambient conditions defining the *site rating*. In cold climates, for example, the gas turbine exhaust temperature would be significantly reduced, resulting in a loss of steam generating capacity; this may require the introduction of supplementary firing in the HRSG. If the design were based on standard day conditions, it could be quite unsatisfactory on a very cold day. The site rating must take into account the seasonal variation of temperature and ensure that the critical requirements can be met. The off-design performance of gas turbines will be discussed in Chapter 9.

The overall efficiency of a cogeneration plant may be defined as the sum of the net work and useful heat output per unit mass flow, divided by the product of the fuel/air ratio and $Q_{\text{net},p}$ of the fuel. For the purpose of preliminary cycle calculations it is sufficiently accurate to evaluate the useful heat output per unit mass flow as $c_{pg}(T_{\text{in}} - 393)$ bearing in mind the need to keep the stack temperature from falling much below 120°C . T_{in} would be the temperature at entry to the HRSG and would normally be the exhaust gas temperature of the turbine. It is unlikely that a regenerative cycle would be used in cogeneration plant because the low exhaust temperature would limit steam production.

Repowering

Another way in which gas turbines can be used with steam turbines is the *repowering* of old power stations. Steam turbines have much longer lives than boilers, so it is possible to create an inexpensive combined cycle plant by replacing an existing fired boiler with a gas turbine and HRSG. The steam conditions for which power stations were designed in the 1950s enable a good match to be achieved between the steam and gas cycles. The power output can be trebled, while the thermal efficiency is increased from about 25 per cent to better than 40 per cent. The result is a vastly superior power plant installed on an existing site. The problems, and cost, of finding a suitable location for a new power station are avoided, and this contributes to the reduction in cost of electricity sent out. A typical repowering project is described in Ref. (7).

An interesting example of repowering is the conversion of an uncompleted nuclear plant to a base-load combined cycle plant, using the steam turbines and generators of the existing scheme but replacing the nuclear reactor as a heat source by 12 85 MW gas turbines and HRSGs. The net result in this instance is a total output of 1380 MW, plus a substantial supply of steam to a chemical plant.

2.6 Closed-cycle gas turbines

The main features of the closed-cycle gas turbine have been described in section 1.3, where it was suggested that a monatomic gas would have advantages over air as the working fluid. Although, as we pointed out, closed-cycle gas turbines are unlikely to be widely used, it will be instructive to quantify some of these advantages. To this end, the performance of a particular closed cycle will be evaluated for air and helium.

Figure 2.30 is a sketch of the plant, annotated with some of the assumed operating conditions, namely LP compressor inlet temperature and pressure, HP compressor inlet temperature, and turbine inlet temperature. The compressor inlet temperatures would be fixed by the temperature of the available cooling water and the required temperature difference between water and gas. A modest turbine inlet temperature of 1100 K has been chosen. (This might have been achievable in the core of the high-temperature nuclear reactor (HTR) discussed in section 1.3.) The higher we can make the pressure in the circuit, the smaller the plant becomes, and we shall assume that a minimum pressure of 20 bar is practicable. Typical values of component efficiencies and pressure loss are: $\eta_{\infty c} = 0.89$, $\eta_{\infty t} = 0.88$, heat-exchanger effectiveness 0.7; pressure loss as percentage of component inlet pressure: in precooler and intercooler 1.0 per cent each, in hot and cold sides of heat-exchanger 2.5 per cent each, in gas heater 3 per cent.

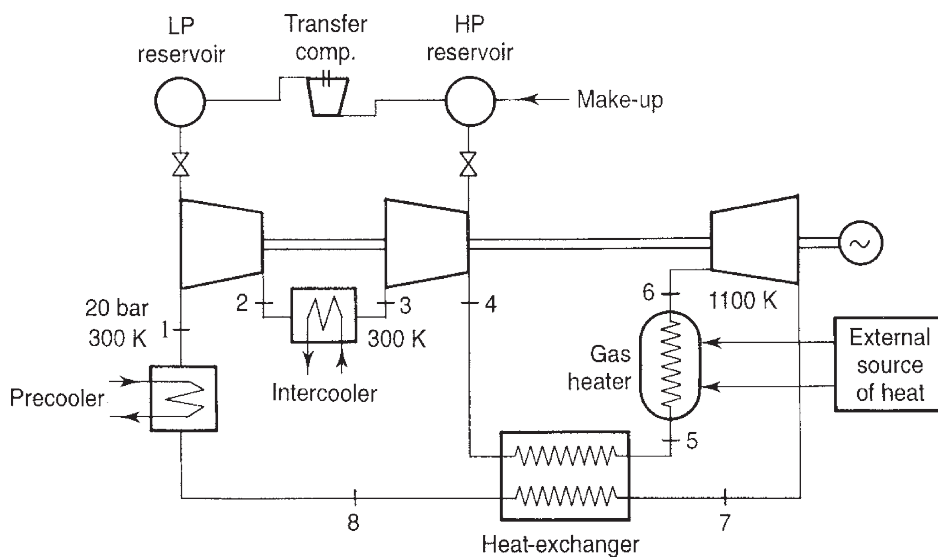


FIG. 2.30 Example of a closed-cycle plant

The performance will be evaluated over a range of compressor pressure ratio p_{04}/p_{01} . It will be assumed that the compression is split between LP and HP compressors such that $p_{02}/p_{01} = (p_{04}/p_{01})^{0.5}$ which leads to an approximately equal division of work input. The usual values of $\gamma = 1.4$ and $c_p = 1.005$ kJ/kg K are assumed to hold throughout the cycle when air is the working fluid, and for helium $\gamma = 1.666$ and $c_p = 5.193$ kJ/kg K. The cycle efficiency is given by

$$\eta = \frac{[(T_{06} - T_{07}) - (T_{04} - T_{03}) - (T_{02} - T_{01})]}{(T_{06} - T_{05})}$$

When comparing the specific work output of cycles using fluids of different density, it is more useful to express it in terms of output per unit volume flow than per unit mass flow, because the size of plant is determined by the former. Here we shall evaluate the specific output from the product of the output per unit mass flow and the density at inlet to the compressor where $T = 300$ K and $p = 20$ bar. The density of air at this state is 23.23 kg/m³ whereas for helium (molar mass 4 kg/kmol) it is only 3.207 kg/m³.

Figure 2.31 shows the results of the calculations and we will consider first the efficiency curves. They suggest that the helium cycle has a slightly lower efficiency. In fact more accurate calculations, allowing for the variation of c_p and γ with temperature in the case of air, lead to almost identical maximum efficiencies. (With helium, c_p and γ do not vary significantly with temperature over the range of interest.) As will be shown below, however, the heat transfer characteristics of helium are better than those of air so that it is probable that a higher heat-exchanger effectiveness can be used with helium without making the heat-exchanger excessively large. The portion of the dashed curve in Fig. 2.31 indicates the benefit obtained by raising the effectiveness from 0.7 to 0.8 for the helium cycle: the efficiency is then 39.5 per cent compared with about 38 per cent for the air cycle.

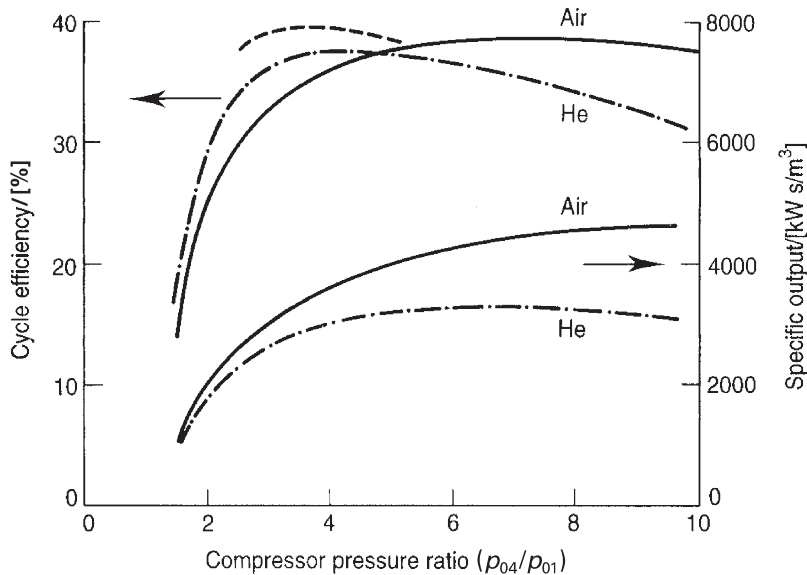


FIG. 2.31 Comparison of air and helium closed-cycle performance

That the heat transfer characteristics are better for helium can be deduced quite simply from the accepted correlation for heat transfer in turbulent flow in tubes [Ref. (1)], namely

$$Nu = 0.023 Re^{0.8} Pr^{0.4}$$

where Nu is the Nusselt number (hd/k). The Prandtl number ($c_p\mu/k$) is approximately the same for air and helium, so that the heat transfer coefficients will be in the ratio

$$\frac{h_h}{h_a} = \left(\frac{Re_h}{Re_a}\right)^{0.8} \left(\frac{k_h}{k_a}\right) \quad (2.24)$$

Suffixes h and a refer to helium and air respectively. From tables of properties, e.g. Ref. (8), it may be seen that the thermal conductivities vary with temperature, but the ratio k_h/k_a is of the order of 5 over the relevant range of temperature. The Reynolds number ($\rho Cd/\mu$) is a function of velocity as well as properties of the fluid, and the flow velocity in the heat-exchanger (and coolers) is determined by the pressure drop entailed. As may be found in any text on fluid mechanics, or Ref. (1), the pressure loss for turbulent flow in tubes is given by

$$\Delta p = \left(\frac{4L}{d}\right)\tau = \left(\frac{4L}{d}\right)f\left(\frac{\rho C^2}{2}\right)$$

where τ is the wall shear stress and f the friction factor given by the Blasius law

$$f = \frac{0.0791}{Re^{0.25}} = \frac{0.0791}{(\rho Cd/\mu)^{0.25}}$$

It follows that for equal pressure losses,

$$(\rho^{0.75} C^{1.75} \mu^{0.25})_h = (\rho^{0.75} C^{1.75} \mu^{0.25})_a$$

ρ_h/ρ_a is 0.138 and μ_h/μ_a is approximately 1.10, so that

$$\frac{C_h}{C_a} = \left(\frac{1}{0.138^{0.75} \times 1.10^{0.25}}\right)^{1/1.75} = 2.3$$

Thus for similar pressure losses the flow velocity of helium can be double that of air.

Using this result we have

$$\frac{Re_h}{Re_a} = \frac{\rho_h C_h \mu_h}{\rho_a C_a \mu_a} = \frac{0.138 \times 2.3}{1.10} = 0.29$$

Finally, from equation (2.24),

$$\frac{h_h}{h_a} = 0.29^{0.8} \times 5 = 1.86$$

and the heat transfer coefficient for helium is therefore almost twice that for air. This implies that the heat-exchanger need have only half the surface area of tubing

for the same temperature difference, or that a higher effectiveness can be used economically.

Turning now to the specific work output curves of Fig. 2.31, from which the comparative size of plant can be deduced, it will be appropriate to make the comparison at the pressure ratio yielding maximum efficiency in each case, i.e. around 4 for helium and 7 for air. Evidently about 45 per cent greater volume flow is required with helium than with air for a given power output. The reason is that under conditions of maximum efficiency, in each case the compressor temperature rises and turbine temperature drops are similar for helium and air, so that the specific outputs on a *mass* flow basis are in the ratio $c_{ph}/c_{pa} \approx 5$. The density of helium is only 0.138 times that of air, however, so that on a volume flow basis the specific output with helium is about 0.7 of that with air.

The increase in size of the heat-exchanger and coolers due to this higher volume flow will be more than offset by the reduction arising from the better heat transfer coefficients and the higher flow velocities that can be used with helium. Higher velocities will also reduce the diameter of the ducts connecting the components of the plant. A 45 per cent increase in volume flow with a doubling of the velocity would imply a 15 per cent reduction in diameter.

The effect of using helium instead of air on the size of the turbo-machinery will only be fully appreciated after the chapters on compressors and turbines have been studied. We have already noted that the work done per unit mass flow in these components is about five times as large with helium as with air. One might expect that five times the number of stages of blading will be required. This is not so, however, because the Mach number limitation on flow velocity and peripheral speed (which determine the work that can be done in each stage) is virtually removed when helium is the working fluid. The reason is that the sonic velocity, $(\gamma RT)^{0.5}$, is much higher in helium. The gas constant is inversely proportional to the molar mass so that the ratio of sonic velocities at any given temperature becomes

$$\frac{a_h}{a_a} = \left(\frac{\gamma_h R_h}{\gamma_a R_a} \right)^{0.5} = \left(\frac{1.66 \times 29}{1.40 \times 4} \right)^{0.5} = 2.94$$

The work per stage of blading can probably be quadrupled with helium, so that the number of stages is only increased in the ratio 5/4. On top of this we have a reduction in annulus area, and therefore height of blading, consequent upon the higher flow velocities through the turbomachinery. Only a detailed study would show whether the turbomachinery will in fact be much larger for the helium cycle.

Because helium is a relatively scarce resource, the decision to use it in a large power plant will probably rest not so much on thermodynamic considerations as the satisfactory solution of the practical problem of sealing the system at the high pressures required. Experience with sealing HP carbon dioxide in gas-cooled reactors has shown that leakage is substantial, and helium, being a lighter gas, will be even more difficult to contain. Although in section 1.7 it was stated that closed-cycle gas turbines are unlikely to be used in nuclear power plant, in-depth studies of the technological requirements for future applications continued until the mid-1990s [Ref. (9)].

NOMENCLATURE

The most widely used symbols in the book are introduced here and they will not be repeated in the lists at the end of other chapters.

C	velocity
c_p	specific heat at constant pressure
f	fuel/air ratio by weight
h	specific enthalpy
ΔH	enthalpy of reaction
M	molecular weight, Mach number
m	mass flow
n	polytropic index
p	absolute pressure
Q	heat transfer per unit mass flow
$Q_{\text{net},l}$	net calorific value at constant p
R, R	specific, molar (universal), gas constant
r	pressure ratio
s	specific entropy
T	absolute temperature
t	temperature ratio
W	specific work (power) output
γ	ratio of specific heats
η	efficiency
ρ	density

Suffixes

0	stagnation value
1, 2, 3, etc.	reference planes
∞	polytropic
a	ambient, air
b	combustion chamber
c	compressor
f	fuel
g	gas
h	heat-exchanger
i	intake, mixture constituent
m	mechanical
N	net
s	stage
t	turbine